

Subject: Adv. Java

Date: 11/11/24

Q. 1. Sec - A ($5 \times 2 = 10$)

1. What are the life cycle methods for a Servlet?

Ans. There are three life cycle methods of a Servlet are -

- `init()` :- Initializes the Servlet.
- `service()` :- Processes requests and responds to clients.
- `destroy()` :- Destroys the Servlet and its resources.

2. What is difference between Get and Post method?

Ans. Feature	Get	Post
1. Purpose	Retrieve data (safe)	Submit data
2. Data location	In the URL	In the request body
3. Data size	Limited by URL length	No size limit
4. Security	Less secure	More secure
5. Caching	Can be cached	Cannot be cached

2. What is difference betw. `GenericServlet` and `HTTPServlet`?

Ans. Feature	Generic Servlet	HTTPServlet
1. Protocol Support	Protocol - independent	Specially for HTTP
2. Methods to Override	<code>service()</code> (Servlet Request req., <code>ServletResponse res</code>)	<code>doGet()</code> , <code>doPost()</code> , <code>doPut()</code> , etc.
3. ease to use	Requires more custom handling of requests	easier for HTTP-based applications

4. What is the purpose of `RequestDispatcher` Interface?

Ans. The `RequestDispatcher` interface in Java EE is used to send requests from a client to a server resource, such as a Servlet, HTML File, or JSP File. The Servlet container creates the `RequestDispatcher` object, which acts as a wrapper around the server resource.

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5. What is the use of welcome-file-list?

Ans: The <welcome-file-list> element in web.xml allows you to define a list of default files that the server will access when a user accesses a directory without specifying a file name.

Section-B (3X7=21)

1. What is difference betⁿ ServletConfig and ServletContext?

Ans

ServletConfig

ServletContext

- ServletConfig object is one per servlet class.

ServletContext object is global to entire web application.

- Object of ServletConfig will be created during initialization process of the servlet.

- Object of ServletContext will be created at the time of web application deployment.

- Scope - As long as a servlet is existing, ServletConfig object will be available, it will be destroyed once the servlet execution is completed.

- Scope - As long as web application is existing, ServletContext object will be available for all servlets and JSP's and it will be destroyed once the application is removed from the server.

- The ServletConfig object is public to a particular servlet only.

- The ServletContext object is public to all servlets and JSP's in a web application.

- It access via getServletConfig() method in the Servlet.

- It access via getServletContext() method in the Servlet.

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2. What is Session Tracking in Servlet? what are cookies? explain some methods used in cookies.

Ans. Session tracking is the mechanism that allows web applications to maintain state across multiple requests from the same client. In stateless protocols like HTTP, each request from a client to a server is independent, meaning the server has no memory of previous requests from the same client. This makes it difficult to maintain user-specific data across requests like user authentication or shopping cart information.

- Cookies are small pieces of data stored by the client (in the user browser) that can be sent back to the server with each HTTP request. Cookies can store stateful information across multiple requests, enabling session tracking or personalization of content.

- Methods of the Cookie class in Java:

1. cookie (string name, string value)
2. setMaxAge (int expiry)
3. setPath (string path)
4. setDomain (string domain)
5. setSecure (boolean flag)
6. setHttpOnly (boolean flag)
7. getName ()
8. getValue ()
9. getMaxAge ()
10. getPath ()

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11. getMain()
12. delete()

2. WAP using servlet to create and display context parameters.
 Ans. Define context parameters in web.xml.

```
<web-app xmlns = "http://java.sun.com/xml/ns/javaee">
  <display-name>ServletContextParamExample </display-name>

  <context-param>
    <param-name>appVersion </param-name>
    <param-value> 1.0.0 </param-value>
  </context-param>

  <context-param>
    <param-name>companyName </param-name>
    <param-value> Tech Corp </param-value>
  </context-param>

  <servlet>
    <servlet-name> ContextParamServlet </servlet-name>
    <servlet-class> com.example.servlet.ContextParamServlet </servlet-class>

    </servlet>

    <servlet-name> ContextParamServlet </servlet-name>
    <url-pattern> display-contextParam </url-pattern>
  </servlet-mapping>

</web-app>
```

Create a servlet to display context parameters.

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```

package com.example.servlet;
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

@WebServlet("/displayContextParams")
public class ContextParamsServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        ServletContext context = getServletContext();
        String appVersion = context.getInitParameter("appVersion");
        String companyName = context.getInitParameter("companyName");

        out.println("<html>");
        out.println("<head><title>Servlet Context Parameters / title</head>");

        out.println("<body>");
        out.println("<h1>Context Parameters from web.xml </h1>");

        out.println("<p><strong>AppVersion</strong> " + appVersion + "</p>");

        out.println("</body>");
        out.println("</html>");
    }
}

```

• Deploy & test :- <http://localhost:8080/your-webapp/displayContextParams>

O/P: Servlet Context Parameters
 AppVersion : 1.0.0
 company name : TeenCorp

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- Sec-C (3X11=33)

1. Write a servlet program to insert a record into the database and display "Record inserted." if record is inserted successfully or "Can't insert a record." message if unsuccessful. The database is Empdb with fields name, employee and first as empname, name, designation, phone.

Ans

```
import java.util.*;
import java.sql.*;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class InsertEmployee extends HttpServlet {
    private static final String JDBC_DRIVER = "com.mysql.cj.jdbc.Driver";
    private static final String DB_URL = "jdbc:mysql://localhost:3306/empdb";
    private static final String DB_USER = "root";
    private static final String DB_PASSWORD = "password";

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        int empno = Integer.parseInt(request.getParameter("empno"));
        String name = request.getParameter("name");
        String designation = request.getParameter("designation");
        String phone = request.getParameter("phone");
        Connection con = null;
        PreparedStatement stmt = null;
        PrintWriter out = response.getWriter();
        try {
```

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```

class.forName("DB-DRIVER");
conn = DriverManager.getConnection(DB-URL, DB-USER,
                                     DB-PASSWORD);

String sql = "Insert into employee (empno, name,
                                     designation, phone) values (?,?,?,?)";
stmt = conn.prepareStatement(sql);
stmt.setInt(1, empno);
stmt.setString(2, name);
stmt.setString(3, designation);
stmt.setString(4, phone);
int rowAffected = stmt.executeUpdate();
if (rowAffected > 0) {
    out.println("<h3> Record inserted successfully </h3>");
}
else {
    out.println("<h3> Can't insert the Record </h3>");
}
catch (Exception e) {
    e.printStackTrace();
    out.println("<h3> Can't insert the record due to an error: </h3>");
}
}
}

```

Q2. Write a Servlet program to demonstrate the redirection by using two methods include and forward.

Ans. index.html

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```

<form action = "Servlet" method = "post">
  Name : <input type = "text" name = "username" /> <br />
  Password : <input type = "password" name = "user pass" /> <br />
  <input type = "submit" value = "login" />
</form>

```

Login.java

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class Login extends HttpServlet {
    public void doPost (HttpServletRequest request, HttpServletResponse
        response) throws ServletException, IOException {
        response.setContentType ("text/html");
        PrintWriter out = response.getWriter();

        String n = request.getParameter ("username");
        String p = request.getParameter ("user pass");
        if (p.equals ("Servlet")) {
            RequestDispatcher rd = request.getRequestDispatcher ("Servlet");
            rd.forward (request, response);
        }
        else {
            out.print ("Sorry, username or password Error!");
            RequestDispatcher rd = request.getRequestDispatcher
                ("/index.html");
            rd.include (request, response);
        }
    }
}

```

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```

WelcomeServlet.java
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class WelcomeServlet extends HttpServlet {
    public void doPost (HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String n = request.getParameter("username");
        out.print("Welcome " + n);
    }
}

```

3. Write a Servlet program to write a cookie in user's computer. Cookie name should be "sessionid" and cookie value should be session's id.

Ans

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class SessionCookie extends HttpServlet {
    protected void doGet (HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
        response.setContentType("text/html");
        HttpSession session = request.getSession(true);

        String sessionId = session.getId();
        Cookie sessionCookie = new Cookie("sessionid", sessionId);
    }
}

```

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```

SessionCookie.setMaxAge (60 * 60);
response.addCookie (sessionCookie);
PrintWriter out = response.getWriter();
out.println("<html><body>");
out.println("<h1> session id cookie created </h1>");
out.println("<p> session id: " + sessionId + "</p>");
out.println("<p> A cookie named 'sessionid' has been created  
and stored in your browser. </p>");

```

```

out.println("</body> </html>");
}

```

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